

Week-4

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Project Summary

This project successfully demonstrates the creation of an end-to-end data pipeline within Microsoft Azure. The implementation was executed in the following steps:

1. **Resource Provisioning:** A dedicated Resource Group was created to logically organize the project assets.
2. **Storage Setup:** An Azure Storage Account was deployed, featuring two Blob Containers (input-data as the source and output-data as the destination). A sample Superstore CSV dataset was uploaded to the source container.
3. **Data Factory Configuration:** An Azure Data Factory instance was provisioned. Secure access was established by creating a Linked Service to the Storage Account utilizing a System-assigned Managed Identity.
4. **IAM Role Assignment:** The Data Factory's Managed Identity was granted the "Storage Blob Data Contributor" role at the Storage Account level to ensure authorized read/write access.
5. **Pipeline Development:** Source and destination datasets were defined. A pipeline was built using a "Get Metadata" activity to validate the presence of the source file, which successfully triggered a "Copy Data" activity.
6. **Execution:** The pipeline was executed in Debug mode, successfully passing metadata validation and transferring the data from the source to the destination container.

Task 1: Explore Azure Portal

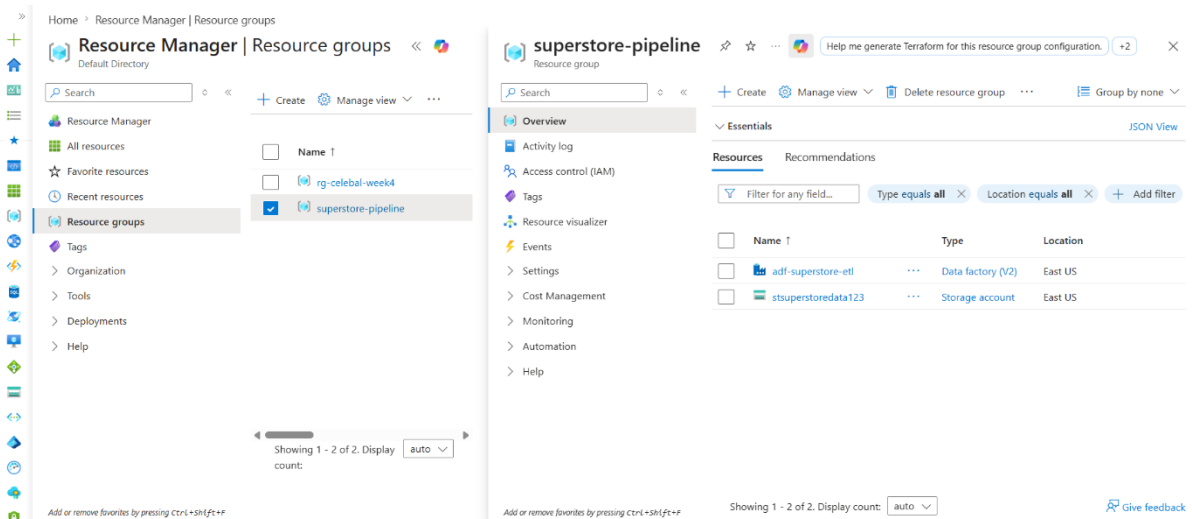


Fig 1: Screenshot showing the **successful creation of the superstore-pipeline** Resource Group in the Azure Portal, which acts as the logical container for this project.

Task 2: Storage Setup

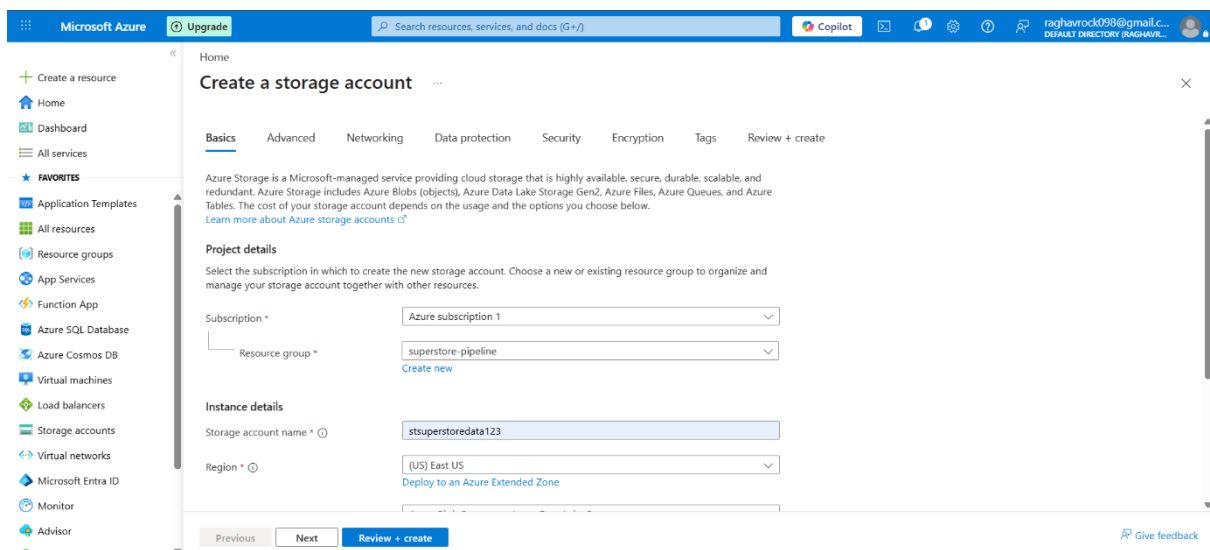


Fig 2: Screenshot showing the **successful creation and listing of the (stsuperstoredata123)** Resource Group in the Azure Portal, which serves as the foundational logical container for all subsequent pipeline resources.

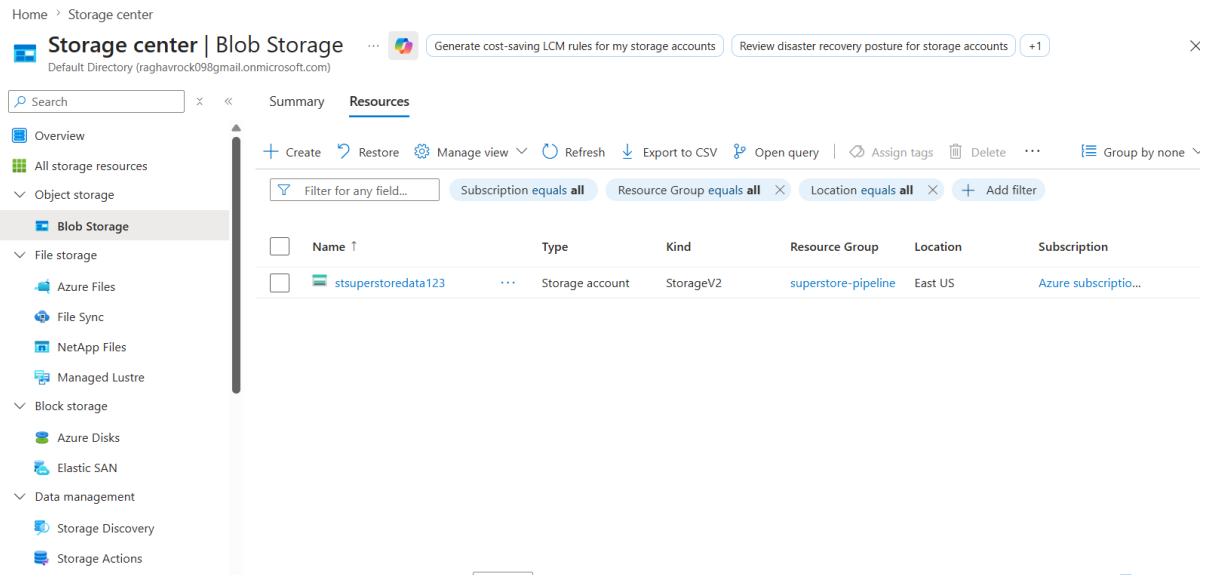


Fig 3: Successful creation and deployment of the Azure Storage Account (stsuperstoredata123) configured for Blob Storage within the designated resource group.

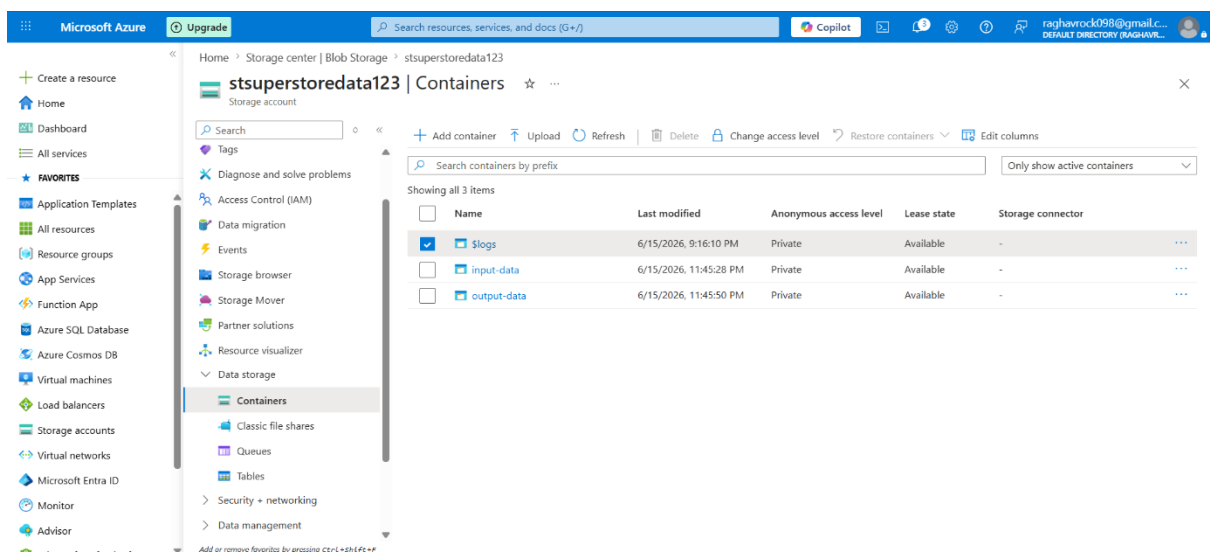


Fig 4: Displaying the successful creation of the input-data (source) and output-data (destination) Blob Containers within the Azure Storage Account, establishing the locations for data ingestion and pipeline output.

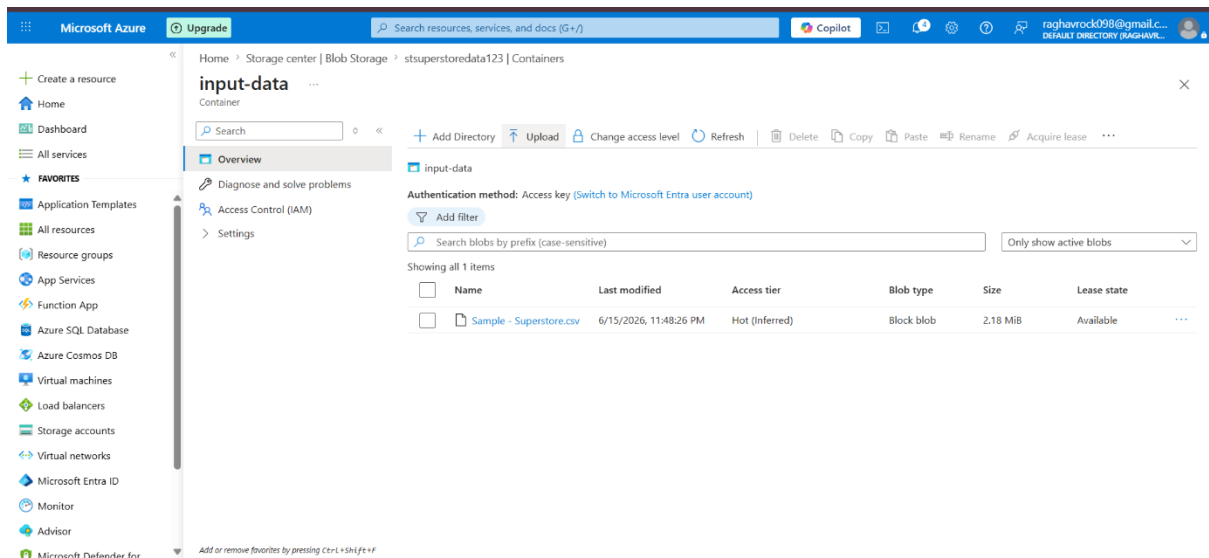


Fig 5: Screenshot verifying the successful upload of the raw Sample- Superstore.csv dataset into the input-data source container in Blob Storage.

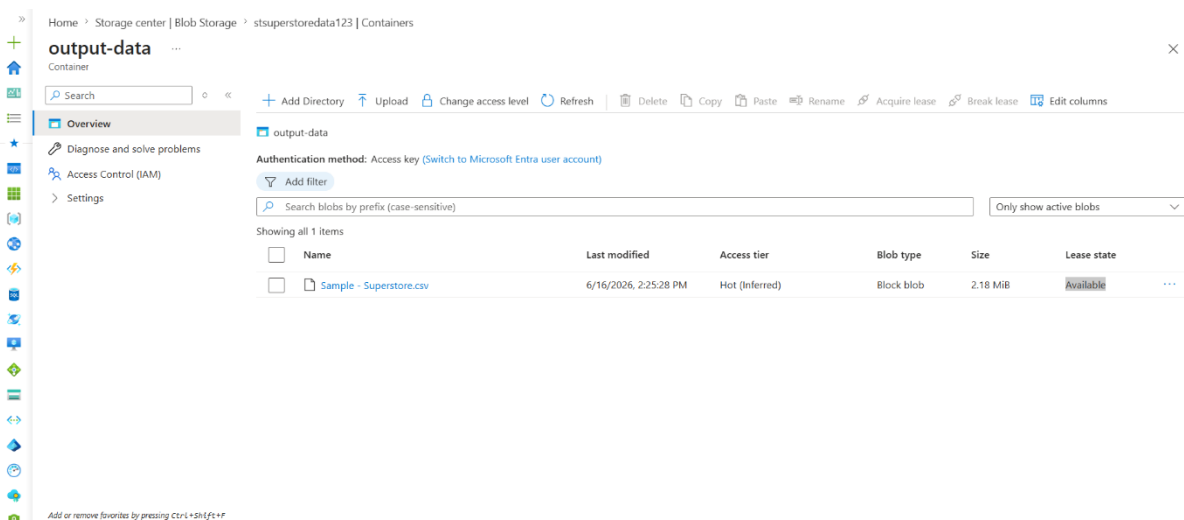


Fig 6: Screenshot of the output-data destination container, proving that the Azure Data Factory pipeline successfully executed and copied the CSV file to the target location.

Task 3: ADF Basics

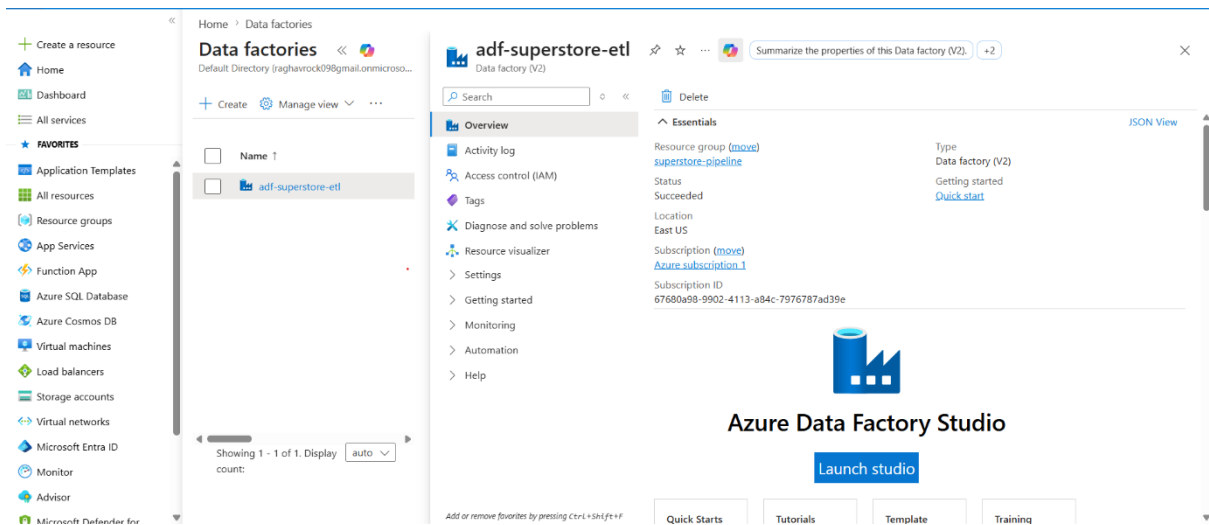


Fig 7: The Azure portal overview page confirming the successful creation and deployment of the Azure Data Factory resource (adf-superstore-etl) within the designated resource group, ready for pipeline development.

Task 3: ADF Basics (Linked Service)

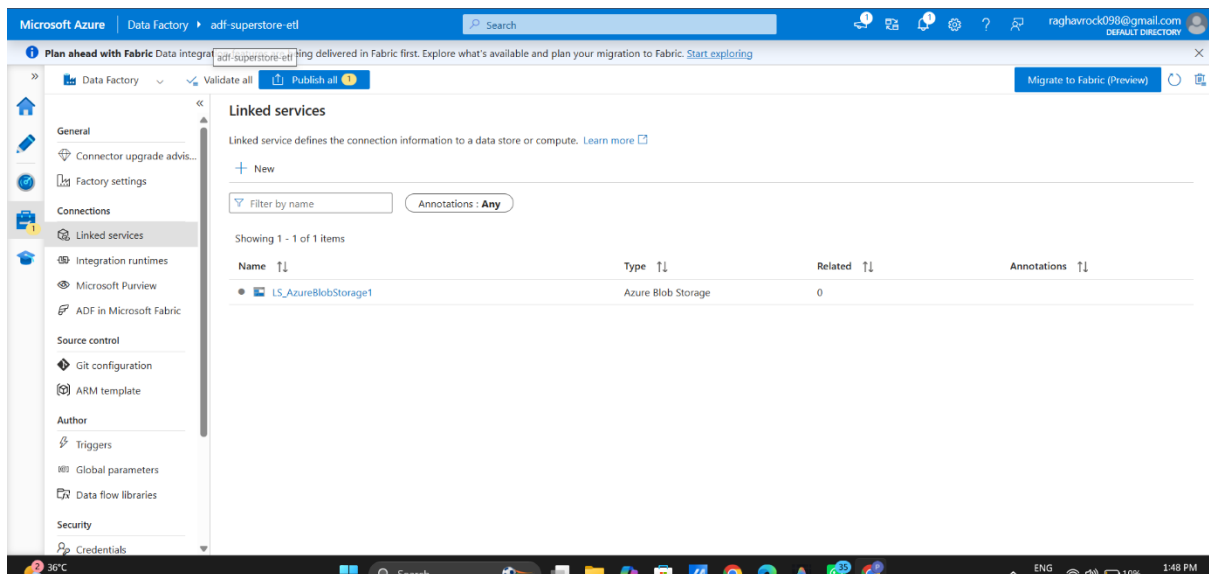


Fig 8: The Linked Service (LS_AzureBlobStorage1) is configured within Azure Data Factory to establish a secure, authenticated connection to the underlying Blob Storage account, utilizing a System-assigned Managed Identity for access control.

Task 3: ADF Basics (Datasets)

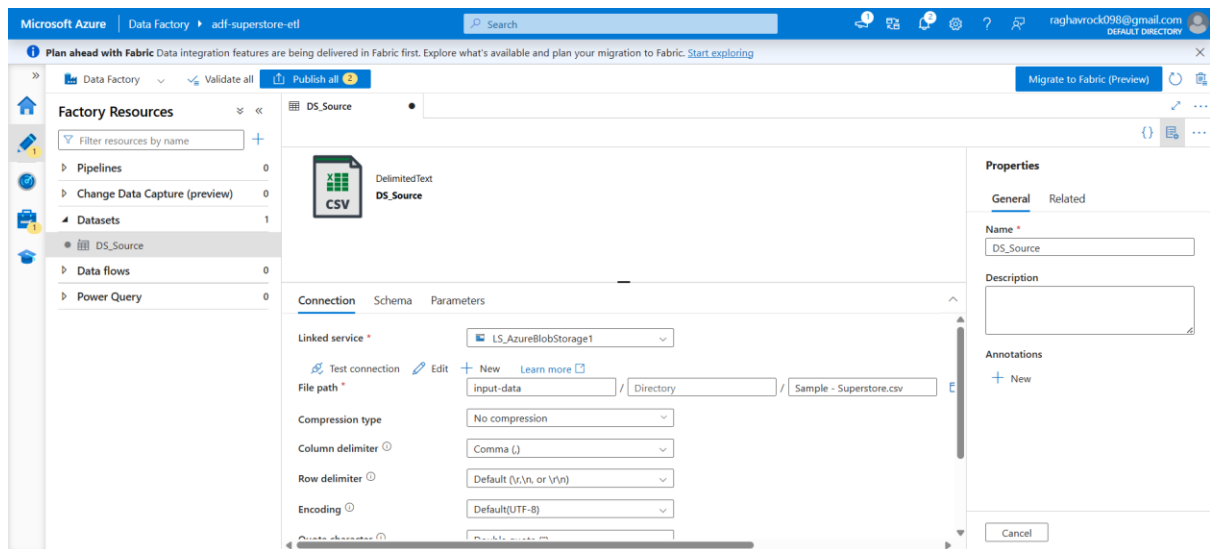


Fig 9: The source dataset (DS_Source) is actively configured to read the incoming Sample-Superstore.csv file directly from the input-data Blob container.

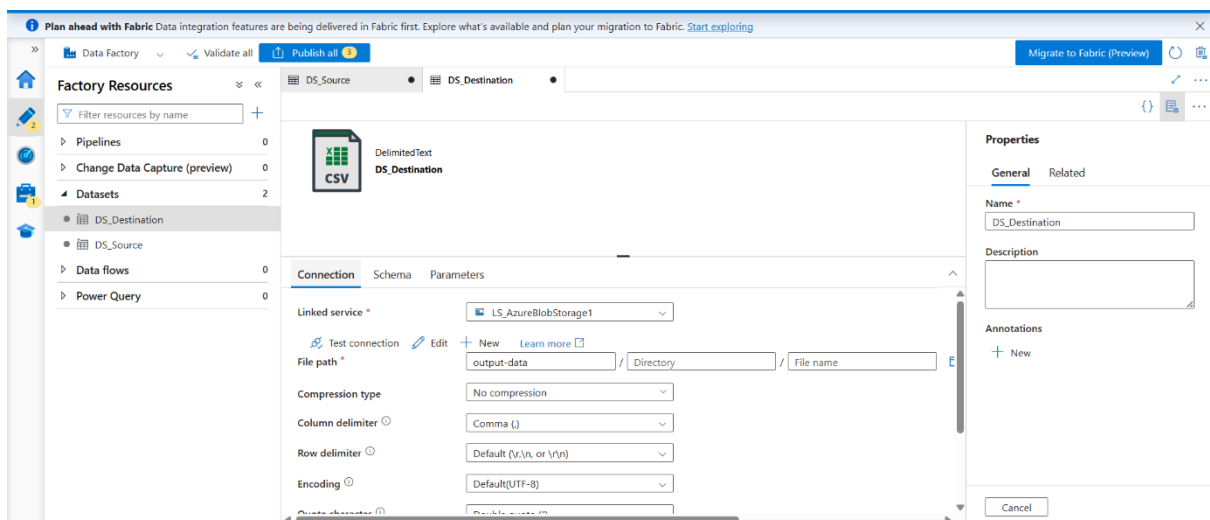


Fig 10: The destination dataset (DS_Destination) is mapped to the output-data Blob container, establishing the exact target location where the pipeline will write the final data.

Task 3: ADF Basics (Get Metadata)

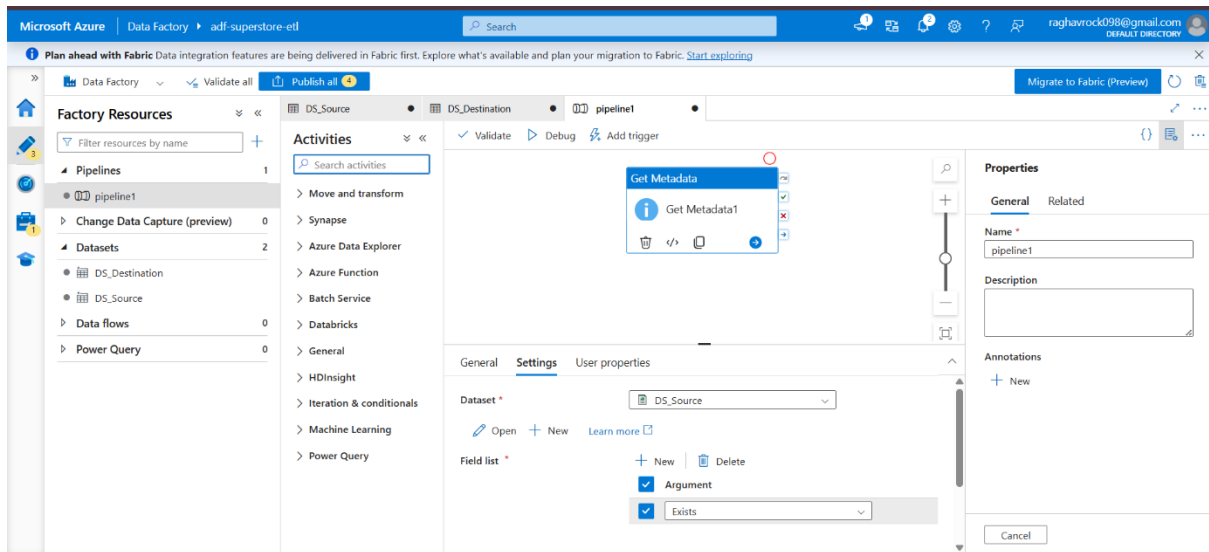


Fig 11: The "Get Metadata" pipeline activity is configured with the "Exists" argument, allowing it to validate the presence of the source file in Blob Storage before any data movement occurs.

Task 4: Pipeline Development

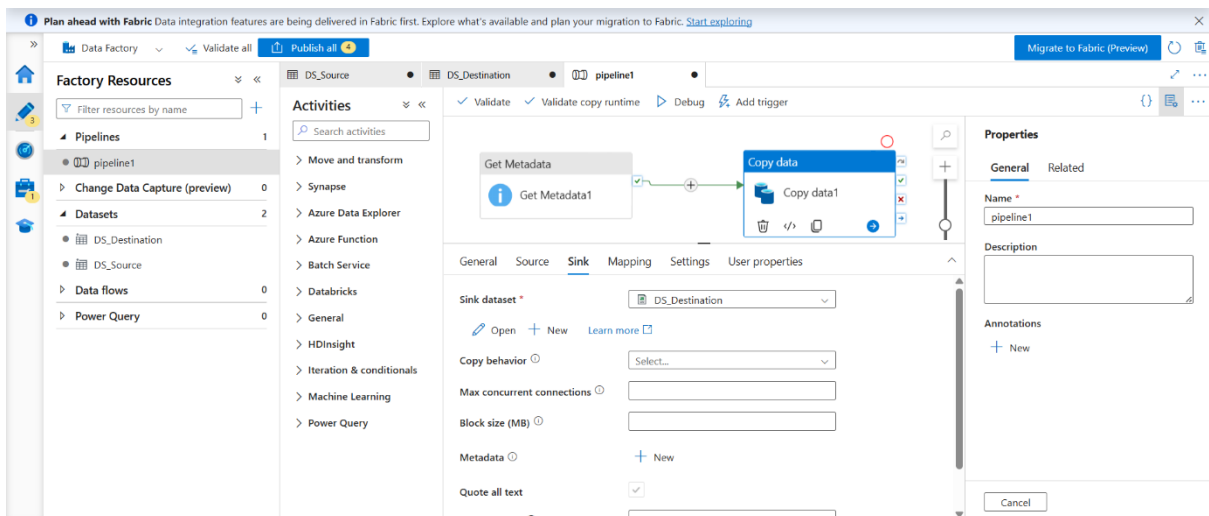


Fig 12: The complete pipeline design features a logical workflow where the "Get Metadata" validation step is connected via a success condition to the "Copy Data" activity, which is configured to write the final output to the destination dataset.

Task 5: Pipeline Execution

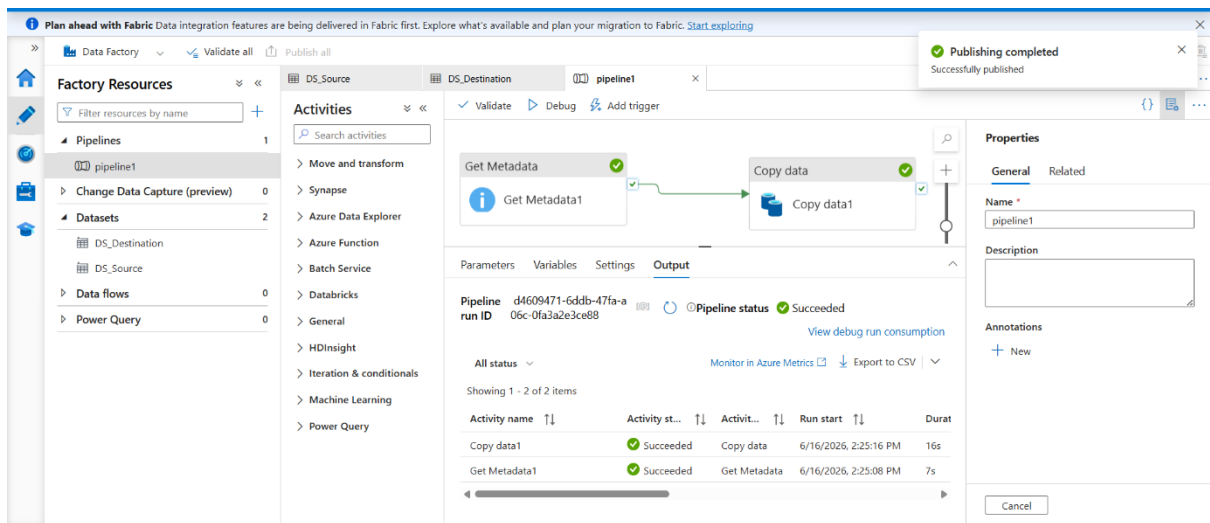


Fig 13: The debug execution output confirms a "Succeeded" status for the overall pipeline, verifying that both the metadata validation and the data copy activities ran flawlessly without any errors.

Task 6: IAM Roles

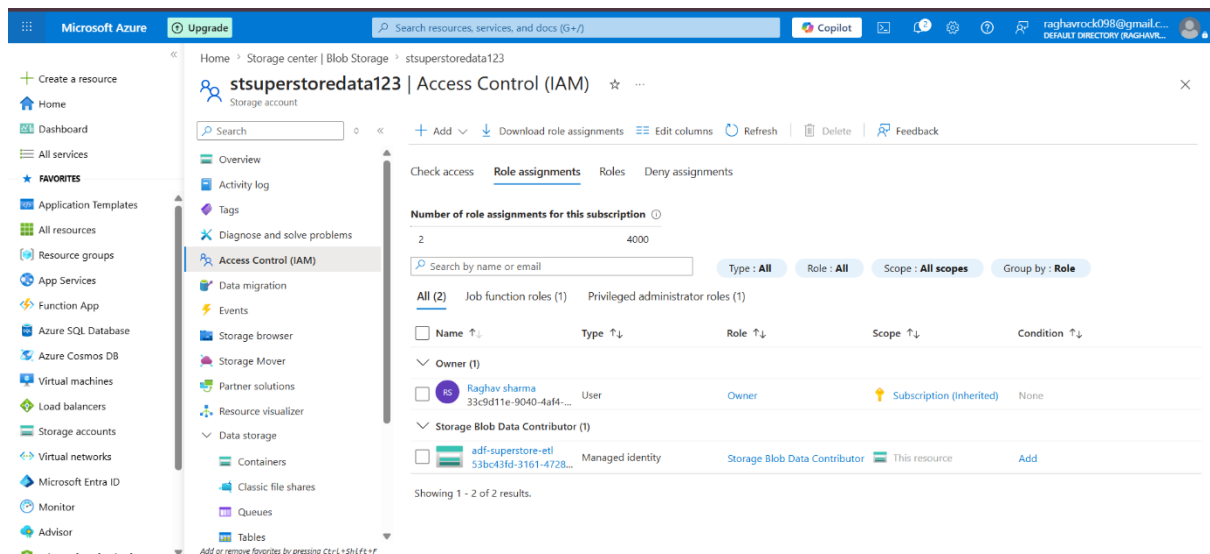


Fig 14: The **Storage Account's Access Control (IAM)** interface verifies that the Azure Data Factory's Managed Identity has been successfully granted the "Storage Blob Data Contributor" role, ensuring secure and authorized data transfer capabilities.